



M/J Grade 7 Digital Discoveries

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADE 7 · ONE SEMESTER · MIDDLE SCHOOL

~4¼ hrs per week · 50 min/day, M–F

About this overview. This document is an example of how the course might be taught. This overview is drawn from the Florida course description and Optima's 2026–27 course plan rather than a fully built course, so scholars may see some variation in tools, projects, and activities from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

In Grade 7 Digital Discoveries, scholars engage with the visible world of computing by using programming, digital creation tools, and introductory AI and immersive VR experiences to explore how hardware, software, and emerging technologies shape communication, problem-solving, and ethical technology use in the modern world. The course treats computer science as a tool for clear thinking and personal expression. Scholars practice breaking problems into manageable parts, reasoning step by step, and explaining how their solutions work, and they consider what it means to use technology responsibly along the way.

THE YEAR AT A GLANCE

1 Computing Around Us (strand)

Scholars look at where computer science shows up in daily life and describe the relationship between hardware and software, including operating systems and the ways people interact with devices and interfaces.

3 Programming with Blocks (strand)

Building programs in a block language using variables, loops, conditionals, and functions. Scholars trace programs, predict outputs, and identify and fix syntax, logic, and runtime errors.

5 Creating and Collaborating (strand)

Scholars use multimedia and design tools to make webpages, digital portfolios, and other artifacts, and work with productivity and collaboration tools while practicing clear team roles.

2 Thinking in Steps (strand)

Computational thinking through decomposition, patterns, and algorithms. Scholars express solutions as a clear series of steps and use logical and arithmetic reasoning with operator precedence.

4 Modeling and Simulation (strand)

Scholars create and modify simulations, use Boolean logic to set project parameters, and explore lists, indexing, and an introduction to the idea of a class in object-oriented programming.

6 Responsible Technology Use (strand)

Digital well-being, online behavior, and the legal and ethical use of information and technology, including the consequences of misuse and a look at technology-related career paths.

A TYPICAL WEEK

- A lesson that introduces the week's computing concept and shows where it appears in everyday life.
- A hands-on coding, building, or digital-creation task that puts the concept to work.
- A short reflection on how the technology works and how it is used responsibly.
- Immersive VR and AI-supported activities woven in where they fit the week's topic.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday, for one semester

- Lessons, coding practice, and projects fill most of the scheduled time.
- Project weeks may run a little longer.

WHAT SCHOLARS NEED

- A computer with reliable internet and access to Canvas.
- Access to Optima's VR campus for immersive activities.
- Microsoft 365 and OneDrive (provided to every scholar for projects and write-ups).

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, when offered live; for 2026–27 M/J Grade 7 Digital Discoveries is listed as an On-Demand course.